

Multi-Source Candidate Data Transformer

Technical Design for the Eightfold Engineering Intern Assignment (Jul/Dec 2026)

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PIPELINE: THE 5-STAGE ETL ARCHITECTURE



- Ingest & Extract:** Source adapters parse raw inputs directly into a shared Intermediate Representation (IR), tagging every field with structural provenance at write-time.
- Sanitize & Canonicalize:** Fields are mapped to strict global standards (E.164, ISO-3166) inline during extraction to prevent downstream poisoning.
- Identity Resolution & Fusion:** Profiles are clustered via an exact email key (with a normalized full-name fallback). Conflicts are resolved via Source Priority Weighting, and corroborating sources boost the overall confidence score.
- Dynamic Projection:** An isolated configuration engine reshapes the canonical record at runtime via JSONPath according to user constraints.
- Validation:** Both the default-schema and custom-projected outputs are strictly validated against JSON Schemas before emission.

CANONICAL SCHEMA & NORMALIZATION

Matches the spec's field list (`candidate_id`, `full_name`, `emails`, `phones`, `location`, `links`, `headline`, `years_experience`, `skills`, `experience`, `education`, `provenance`, `overall_confidence`). Phones → E.164 via `phonenumbers` (default region US, with a 10-digit heuristic for Indian numbers). Dates → YYYY-MM via `dateutil`. Countries → ISO-3166 alpha-2 via `pycountry`, with a manual fallback table for common aliases (usa, uk). Skills are lowercased and passed through a synonym map (js→javascript, k8s→kubernetes, etc.) for canonicalization.

MERGE & CONFLICT RESOLUTION

Profiles are grouped by a match key, primarily an exact email match, with a fallback to normalized lowercase `full_name` when no email is present, so a candidate's CSV and GitHub records can still merge on name alone. This is a deliberate, scoped trade-off: it improves recall for sparse-source candidates at the cost of a small false-merge risk between distinct people who share a name and have no email. This risk is acceptable for a small demo dataset, but we would require a second disambiguator (e.g. company) before trusting it at scale. Within a matched group, each source carries a static reliability weight (CSV: 0.9, GitHub: 0.7); on a field conflict, the higher-weighted source wins. Arrays (emails, phones, skills, experience, provenance) are unioned and deduplicated rather than overwritten, so no data is silently dropped. Confidence is additive. A profile's base confidence is boosted by 20% of each merged source's confidence, capped at 1.0. `years_experience` is derived post-merge by summing deduplicated experience-entry durations.

RUNTIME CONFIG / PROJECTION

The projection layer reads the config's `fields` list and, per field, pulls a value from the canonical record via JSONPath (`from`), applies a requested normalizer (E164, canonical), and resolves missing values per the config's `on_missing` mode (`null` / `omit` / `error`). Confidence and provenance are toggled at the top level via `include_confidence`. The canonical record and projection layer are kept as separate modules so reshaping output never mutates internal state. Both the default-schema and the custom-projected outputs are validated with a JSON Schema validator before being returned: the default path against the internal schema, and the custom path against a schema generated from the request's `fields` list.

EDGE CASES HANDLED

- Missing/garbage source:** extractors wrap per-row parsing in try/except, log a warning, and continue rather than aborting the run.
- Duplicate candidate across sources:** merged via the email/name key, never duplicated in output.
- Conflicting field values:** resolved via source-priority weight, never silently overwritten without a rule.
- Missing required field in projection:** explicit error/omit/null modes per config, never invented.
- Same-name, different-person collision:** acknowledged as an accepted risk under the name-fallback match, not silently ignored.

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- LinkedIn and resume/notes extractors are omitted since CSV and GitHub already satisfy the structured/unstructured requirement.
- Fuzzy/phonetic name matching is excluded; only exact-lowercase match is implemented.
- Per-field validation error messages beyond a single schema-validation failure are omitted.
- Education extraction is omitted as no in-scope source provides this data.